

Failure Cases and Mitigation Plan

1 Writer Robot Failure

Failure

- Writer becomes immobilized, loses power, suffers hardware failure or becomes unable to continue exploration.

Consequence

- The Writer can no longer explore or transmit newly acquired information.

Mitigation

- Store important information progressively rather than keeping it only on the Writer.
- Transfer event and spatial information to nearby beacons during exploration.
- Use multiple Writers to prevent a single robot failure from stopping the entire mission.
- Allow another Writer or Executor to use the information already preserved.

Validation

- Disable a Writer during exploration.
- Verify that previously transmitted information remains accessible.
- Verify that another robot can continue from the preserved information.

2 Beacon Failure

Failure

- Beacon loses power, communication or hardware functionality.

Consequence

- Part of the communication chain may become unavailable.

Mitigation

- Maintain overlapping communication coverage where possible.
- Detect missing acknowledgements or degraded communication.
- Deploy a replacement beacon when required.
- Use neighbouring beacons to preserve previously stored information.

Validation

- Disable individual beacons.
- Measure whether information can still propagate.
- Determine the maximum tolerable number of consecutive failed beacons.

3 ESP-NOW Communication Loss

Failure

- Excessive distance, obstruction, interference or node failure causes packet loss.

Consequence

- Information may not reach the next beacon.

Mitigation

- Monitor communication quality.
- Use acknowledgements and retransmissions.
- Deploy additional beacons when communication quality becomes insufficient.
- Keep important information locally until successful transmission is confirmed.

Validation

- Increase distance between nodes.
- Introduce obstacles.
- Measure packet-loss and recovery behaviour.

4 Writer Becomes Disconnected from the Beacon Network

Failure

- Writer moves outside the communication range of the nearest beacon.

Consequence

- Newly generated information cannot immediately be transmitted.

Mitigation

- Maintain a local buffer of unsent events.
- Attempt retransmission when communication is restored.
- Use communication-quality measurements to determine when another beacon should be deployed.
- Avoid discarding events because of temporary communication loss.

5 SLAM / Localization Failure

Failure

- Sensor degradation, repetitive geometry, accumulated error or loss of environmental features causes inaccurate localization.

Consequence

- Map distortion.
- Incorrect event coordinates.
- Incorrect beacon positions.

Mitigation

- Fuse multiple sensors instead of relying on a single measurement source.
- Monitor localization confidence.
- Use beacon references to provide additional spatial constraints.
- Re-localize when sufficient environmental information becomes available.

Validation

- Introduce temporary sensor failures.
- Compare localization error with normal operation.
- Test long tunnel sections and repetitive geometry.

6 Incorrect Frame Translation

Failure

- Incorrect transformation between the local SLAM frame and the global coordinate frame.

Consequence

- Correctly detected events are assigned incorrect global coordinates.

Mitigation

- Establish the entrance/reference position carefully.
- Store the reference transformation in the beacon infrastructure.
- Validate the transformation against known reference points.
- Associate uncertainty with transformed coordinates.

Validation

- Use multiple known points.
- Compare calculated and reference global positions.
- Measure transformation error throughout the explored region.

7 Computer-Vision False Detection

Failure

- Camera detects an object as a victim when no victim exists.

Consequence

- False event is stored and may cause an unnecessary rescue mission.

Mitigation

- Store detection confidence.
- Use additional observations for confirmation.
- Combine computer vision with RF sensing when available.
- Allow later robots to verify uncertain detections.

8 Computer-Vision Missed Detection

Failure

- A victim is not visually detected because of darkness, occlusion, distance or orientation.

Consequence

- Victim remains undetected by the visual system.

Mitigation

- Use Wi-Fi CSI as a complementary sensing modality.
- Allow RF detection to identify regions requiring additional investigation.
- Use different robot viewpoints where possible.

9 CSI False Positive

Failure

- Environmental RF variations are interpreted as human presence.

Consequence

- An unnecessary investigation may be initiated.

Mitigation

- Store detection confidence.
- Establish environmental baselines.
- Validate detections using additional measurements.
- Use computer vision when the region is visually accessible.
- Improve the classifier using representative mine-environment data.

10 CSI False Negative

Failure

- A victim does not produce a sufficiently distinguishable RF signature.

Consequence

- RF sensing fails to detect the victim.

Mitigation

- Keep computer vision as an independent detection modality.
- Use multiple spatially separated CSI nodes.
- Adjust node placement when investigating a suspicious region.
- Treat CSI as complementary rather than the sole victim-detection mechanism.

11 CSI Infrastructure Failure

Failure

- Wi-Fi source, CSI node, aggregator or processing unit becomes unavailable.

Consequence

- RF sensing coverage is reduced or lost.

Mitigation

- Distribute CSI nodes rather than relying on one sensing node.
- Use Writers and/or beacons as distributed Wi-Fi sources.
- Continue communication using the independent ESP-NOW network.
- Fall back to computer vision and conventional sensors.

12 Outside Network Station Failure

Failure

- Station loses power, communication or hardware functionality.

Consequence

- Information from the associated entrance cannot reach the Command Post.

Mitigation

- Use multiple stations where required.
- Provide independent power through solar panels and batteries.
- Maintain local information inside the beacon network.
- Restore communication when the station becomes operational again.

13 Optical-Fibre Failure

Failure

- Fibre connection between the first beacon and Outside Network station is interrupted.

Consequence

- The affected entrance loses its direct connection to the external infrastructure.

Mitigation

- Maintain the mission information inside the beacon network.
- Detect the loss of the external connection.
- Restore the physical link when possible.
- Use another entrance/station when the mission architecture permits it.

14 Robot Battery Depletion

Failure

- Robot reaches insufficient battery capacity before completing its mission.

Consequence

- Robot may become immobilized inside the environment.

Mitigation

- Monitor battery level continuously.
- Include estimated return cost in autonomous decision-making.
- Reserve sufficient energy for returning to a known location.
- Use Outside Network stations for robot recharging between missions.

15 Partial Exploration

Failure

- Writer fails, runs out of energy, loses communication or reaches another operational limitation before completing its assigned region.

Consequence

- The environment remains partially unknown.

Mitigation

- Preserve all information collected before failure.
- Record explored and unexplored regions.
- Allow another Writer to continue from the remaining frontier.
- Use the existing beacon map to avoid unnecessary re-exploration.

16 Executor Cannot Reach Target

Failure

- Target route becomes inaccessible because of an obstacle, map error or environmental change.

Consequence

- Executor cannot complete the mission using the original route.

Mitigation

- Allow local re-planning around newly detected obstacles.
- Use the existing map as a reference rather than treating it as immutable.
- Request/obtain additional information when communication permits.
- Preserve newly discovered information for subsequent missions.

17 Complete Communication Loss

Failure

- Robot temporarily loses communication with the beacon network.

Consequence

- Robot cannot exchange information with the rest of the system.

Mitigation

- Continue operating using locally stored information.
- Buffer newly generated events.

- Attempt to reconnect with the nearest known beacon.
- Return toward the last known communication region if continued operation becomes unsafe.

18 Multiple Simultaneous Failures

Failure

- Several Writers, beacons or sensing nodes fail during the same mission.

Mitigation principle

- Avoid single points of failure wherever practical.
- Store critical information redundantly.
- Separate **sensing**, **communication**, and **spatial-memory** functions.
- Use multiple Writers and multiple entrances.
- Allow the system to degrade progressively rather than stopping after one component failure.

Validation

Conduct progressively harder scenarios:

Single failure → **two simultaneous failures** → **communication + robot failure** → **partial network failure**

and measure whether the mission can still be completed or recovered.